



Robust design of a 2-DOF GMV controller: A direct self-tuning and fuzzy scheduling approach

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ABSTRACT

This paper presents a study on self-tuning control strategies with generalized minimum variance control in a fixed two degree of freedom structure – or simply GMV2DOF – within two adaptive perspectives. One, from the process model point of view, using a recursive least squares estimator algorithm for direct self-tuning design, and another, using a Mamdani fuzzy GMV2DOF parameters scheduling technique based on analytical and physical interpretations from robustness analysis of the system. Both strategies are assessed by simulation and real plants experimentation environments composed of a damped pendulum and an under development wind tunnel from the Department of Automation and Systems of the Federal University of Santa Catarina.

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1. Introduction

Over the past few years, the driving force behind good control strategies with designs based on low-order Linear Time Invariant Systems has been Self-Tuning Control (STC) [1]. Despite all the tools and techniques derived from the Robust Control Theory to deal with Plant-Model-Mismatch (PMM) issues in controller design, there exist ways to adapt a model in order to follow process behavior changes, like online identification, that could diminish PMM so designs made in real time – say it online – become more accurate and every control strategy adopted would benefit from it.

In the sense presented above, this paper aims at hybridization of adaptive control with Generalized Minimum Variance Control (GMVC) [2,3] in two ways: (i) by means of online identification through a Recursive Least Squares (RLS) estimator in order to constantly update directly the controller and the control law, and (ii) by means of scheduling the GMV controller parameters with a Mamdani fuzzy logic controller [4–6] based on robust design criteria.

Both adaptive techniques chosen, in fact, lay in a state of dependence with GMVC to provide good control results. In this way,

the major contributor, well-known for being a simple predictive control structure and an effective controller for plants with time delays, is the GMVC. In this paper, the GMVC is presented within an incremental approach and structured as a two degree of freedom (2-DOF) version of this controller, hereinafter designated as GMV2DOF, which means that only two tuning parameters are available and they consequently confine the GMVC to almost a minimal order controller with ease in the design procedure, since it has a fixed small number of tuning parameters.

The motivation to work with fixed structured GMVC comes from the simplification requirements sometimes needed by the industry for fast commissioning of hundreds of control loops, where plant technicians commonly have only few pieces of information about the process behavior in order to build a consistent design model to implement; for example, a well planned pole-assigned GMVC [7]. Instead, fixed structured GMV controllers, such as those presented by Doi and Mori [8], are more suitable for trial and error pre-tuning of adaptive GMVC, since the principal tuning parameter is related to control signal energy factor constraint, in which the operational range is commonly well-known by plant operators and technicians.

Looking forward to give more freedom of design and better control of operational bandwidth, the 2-DOF structure robustness is evaluated using sensitivity magnitude plots in the frequency domain and comparisons are made to the minimal order GMVC of [8]. It is important, however, to emphasize that GMV2DOF is not a new controller, but only a particular solution of the GMVC proposed by Clarke and Gawthrop [2], and similar practices of 2-DOF particularizations can be found in [9,10].

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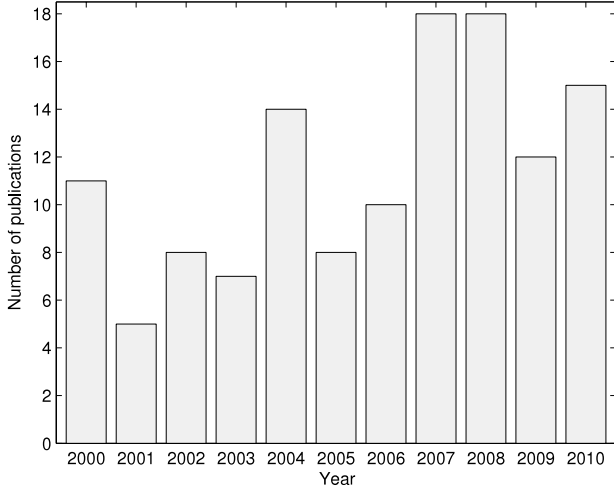


Fig. 1. Number of GMV related publications over the past decade.

It may seem awkward, however, to consider a study on robustness of GMVC at a time where more sophisticated predictive controller techniques are available, but since it was introduced, GMVC is being applied in several application fields, such as: health and medicine [11]; aeronautics [12]; process control [13]; communication systems [14]; power systems [15]; and aerial robotics [16]. It may not have the same impact nowadays as it once had, but it is still being applied and investigated as reference for performance benchmark and process control as depicted by Fig. 1, showing the number of related publications on journals and conferences over the past decade (data from <http://www.scopus.com/>, accessed on February 10, 2011).

Beyond the relevance of past and recent publications mentioned, GMVC is a member of the Model Predictive Control (MPC) family based on infinite horizon optimization [17], well suited and simple to analyze robustness using classical and modern techniques, since the main problem is focused at steady-state condition, whilst more sophisticated MPCs, e.g. receding horizon controllers [17], require sophisticated form of analysis as well. Due to the infinite horizon optimization characteristic of GMVC, the reader should keep in mind that the transitory behavior must be considered somehow in the design procedure, which is done with discretion in this paper while considering the compromise between robustness and performance to retain the control signal constrained to avoid undesired transitory behavior—common practice of any unconstrained optimal design technique is good sense and/or common sense regarding design criteria [18,19].

This paper is organized starting from a brief review of the GMV controller approach by Doi and Mori [8] with robust asymptotic tracking property [20,21] evaluated by the final value theorem. Then, the modified structure proposed in this paper is introduced and evaluated in the same manner and also by mapping GMV2DOF to an RST form [21], in order to take a static snapshot of the system, allowing analytical robustness analysis [22,23]. Subsequently, the STC with RLS estimator development is presented and experimented with a damped pendulum position control and a wind tunnel airstream control [24]. Following up, the STC with the Mamdani Fuzzy GMV2DOF Parameters Scheduling is presented and experimented with both mentioned plants.

2. Incremental GMVC design

Consider the Single-Input Single-Output (SISO) system described by an Auto-Regressive Integrated Moving Average with exogenous Input (ARIMAX) model:

$$A(z^{-1})y(k) = z^{-d}B(z^{-1})u(k) + C(z^{-1})\frac{\xi(k)}{\Delta} \quad (1)$$

$$A(z^{-1}) = 1 + a_1z^{-1} + \dots + a_{n_a}z^{-n_a}$$

$$B(z^{-1}) = b_0 + b_1z^{-1} + \dots + b_{n_b}z^{-n_b}$$

$$C(z^{-1}) = 1 + c_1z^{-1} + \dots + c_{n_c}z^{-n_c}$$

where $\Delta = 1 - z^{-1}$, $y(k)$ is the output, $u(k)$ is the input, d is the discrete time delay and $\xi(k)$ represents a white noise disturbance.

The incremental GMVC control problem is to find the control signal $\Delta u(k)$ which minimizes the cost function $J = \mathbf{E}[\phi^2(k+d)]$, where $\mathbf{E}[\cdot]$ denotes the expectation operator and the generalized output is

$$\phi(k+d) = P(z^{-1})y(k+d) - T(z^{-1})y_r(k+d) + Q(z^{-1})\Delta u(k), \quad (2)$$

$$P(z^{-1}) = 1 + p_1z^{-1} + \dots + p_{n_p}z^{-n_p},$$

$$T(z^{-1}) = t_0 + t_1z^{-1} + \dots + t_{n_t}z^{-n_t},$$

$$Q(z^{-1}) = q_0 + q_1z^{-1} + \dots + q_{n_q}z^{-n_q},$$

with the polynomials $P(z^{-1})$, $T(z^{-1})$, $Q(z^{-1})$, respectively behavior weighting the output, reference and input signals.

By inspection of the right hand side of the generalized output in (2), $y(k+d)$ is a future measurement not available at instant k , so it must be predicted in order to work with $\phi(k+d)$. The Diophantine equation in (3) gives some auxiliary polynomial relations to achieve the prediction:

$$P(z^{-1})C(z^{-1}) = \Delta A(z^{-1})E(z^{-1}) + z^{-d}F(z^{-1}) \quad (3)$$

$$E(z^{-1}) = 1 + e_1z^{-1} + \dots + e_{n_d-1}z^{-(d-1)}$$

$$F(z^{-1}) = f_0 + f_1z^{-1} + \dots + f_{n_f}z^{-n_f}$$

$$n_f = \max(n_p n_c - 1, n_a).$$

Multiplying the ARIMAX model in (1) by $E(z^{-1})$ allows the substitution of the Diophantine equation into the re-arranged ARIMAX model equation. Doing a time-shift of d steps ahead, the needed future output could be predicted with present information – say it $y(k+d|k) = \hat{y}(k+d)$ – the predicted output of the system, which is

$$\hat{y}(k+d) = \frac{B(z^{-1})E(z^{-1})\Delta u(k) + F(z^{-1})y(k)}{P(z^{-1})C(z^{-1})} + \frac{E(z^{-1})}{P(z^{-1})}\xi(k+d). \quad (4)$$

By insertion of (4) into the generalized output in Eq. (2),

$$\phi(k+d) = \frac{BE + CQ}{C}\Delta u(k) + \frac{F}{C}y(k) - Ty_r(k+d) + E\xi(k+d), \quad (5)$$

where all (z^{-1}) were hidden from (5) to (9) for the sake of simplicity. Solving $\partial J / \partial \Delta u(k) = 0$, and taking into account that, since future noise information is not available, the following control law is obtained:

$$\Delta u(k) = \frac{CTy_r(k+d) - Fy(k)}{BE + CQ}. \quad (6)$$

In order to evaluate the servo behavior of the closed-loop system with the final value theorem, the control law in (6) is substituted into the ARIMAX model in (1), resulting in:

$$y(k) = \frac{BT}{PB + \Delta AQ}y_r(k) + \frac{BE + CQ}{PB + \Delta AQ}\xi(k). \quad (7)$$

To guarantee servo mode or reference tracking, the error $e(k) = y_r(k) - y(k)$ in steady-state must be null. Applying the final value theorem over $e(k)$ while considering the unit step as the reference signal, gives:

$$e(k) = \left[z^d C T - \frac{B T}{P B + \Delta A Q} \right] y_r(k) - \left[\frac{B E + C Q}{P B + \Delta A Q} \right] \xi(k)$$

$$e(\infty) = \lim_{z \rightarrow 1} [(1 - z^{-1})e(k)] = C(1)T(1) - \frac{T(1)}{P(1)}. \quad (8)$$

The result obtained in (8) must be taken as a warning when choosing the polynomials $P(z^{-1})$ and $T(z^{-1})$. It forces a restricted condition of $P(1) = T(1)$ [7,8,25] that should be met for reference tracking. The same result is obtained for the case where a load disturbance is sensed at the output.

Now, consider the same problem when a load disturbance is sensed at the input of the system, which means, adding a disturbance signal $d(k)$ to the r.h.s. of (6). This way, a new control signal – contaminated by $d(k)$ – is produced. Repeating what was done to get to (7), now with the disturbed input, the following output is obtained:

$$y(k) = \frac{B T}{P B + \Delta A Q} y_r(k) + \frac{B E + C Q}{P B + \Delta A Q} \xi(k) + \frac{B E + C Q}{C(P B + \Delta A Q)} \Delta d(k). \quad (9)$$

Again, by evaluation of the new $e(k)$ with (9) using the final value theorem for the unit step reference tracking, results in the same restricted condition as in (8). So, despite any modification, if the $P(1) = T(1)$ restricted condition is met, the GMVC incremental approach guarantees reference tracking of step and square wave sequences even when subjected to load disturbances.

For different types of reference sequences, such as sinusoidal, sawtooth and any other ramp-type kind of wave, the GMVC incremental approach will only guarantee that the error will vanish with time, if the variations required by the reference are achievable by the system under consideration. If otherwise, there will be a residual steady-state error due to a delay, d_e , between the reference and the output, which could be compensated by a predictive reference sequence of the kind $y_r(k + d + d_e)$. A simple simulation example is given to illustrate the use of d_e and how incremental GMVC handles different types of references.

2.1. Simple example: incremental GMVC for reference tracking

Consider the following model,

$$Y(s) = \frac{1}{s+1} U(s) \rightarrow y(k) = \frac{0.0952z^{-1}}{1 - 0.9048z^{-1}} u(k), \quad (10)$$

with sampling time $T_s = 0.1$ s, the incremental GMVC is designed in order to find the control law $\Delta u(k)$ using $P(z^{-1}) = T(z^{-1}) = 1$ and $Q(z^{-1}) = 2$ to track a sinusoidal $[y_{r_{\sin}}(k)]$, a sawtooth $[y_{r_{\text{saw}}}(k)]$ and a step $[y_{r_{\text{step}}}(k)]$ wave sequence, compensated with $d_e = 2$ selected by trial and error. The references used in this example can be reconstructed using the following functions:

$$y_{r_{\sin}}(k) = \sin\left(\frac{\pi}{10} k T_s\right) \quad (11)$$

$$y_{r_{\text{saw}}}(k) = \frac{2}{\pi} \sum_{i=1}^{\infty} \frac{\sin\left(\frac{\pi}{10} k T_s i\right)}{i} \quad (12)$$

$$y_{r_{\text{step}}}(k) = \begin{cases} -1, & 800 \leq k < 1000 \\ 1, & k \geq 1000. \end{cases} \quad (13)$$

The results of this simple example can be seen at Fig. 2(a)–(c), where the last one shows the error tending to zero when the system in (10) is tasked to track the references in (11)–(13).

2.2. Dynamic GMV2DOF analysis

The GMV controller incremental approach itself guarantees robust tracking property in steady-state, but how much time a

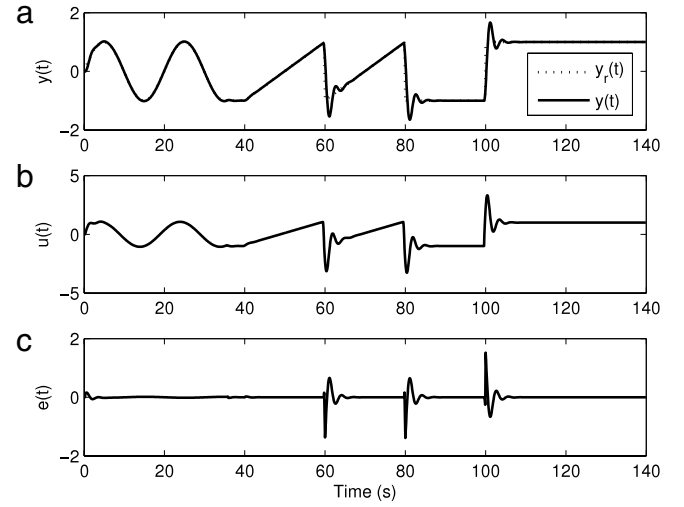


Fig. 2. Results of a simple example on how the incremental GMVC structure can handle step, sinusoidal and ramp-like reference sequences, shown as: (a) reference and output signals; (b) control signal; (c) error signal.

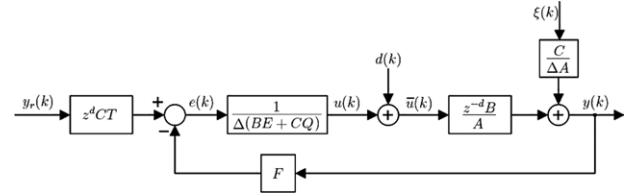


Fig. 3. Block diagram of the control loop.

process could take to get there satisfactorily and inside a restricted and safe scope of operational regions? The answer to this question is very subjective, since every process demands a different kind of treatment, but all of them always need to be kept working respecting some constraint.

The Incremental GMVC structure explored by Doi and Mori [8] was essentially focused on inflicting weight over the control signal by means of a $Q(z^{-1}) = \lambda$ while retaining $P(z^{-1}) = T(z^{-1}) = 1$. The calibration of λ , in that case, was mostly related to energy constraints regarding control effort in order to have good disturbance rejection and reference tracking, but without any consideration being made to how this parameter would affect the relative stability of the control loop. Indeed, λ was heuristically selected.

Before the addition of another parameter to the Incremental GMVC $Q(z^{-1})$ polynomial and introduce GMV2DOF, first, the Doi and Mori [8] case is analyzed analytically and then, the work is expanded to the case with the modified $Q(z^{-1})$, where

$$Q(z^{-1}) = \frac{\lambda}{1 - \gamma z^{-1}}. \quad (14)$$

One way to have a better viewpoint of the control loop is to represent it in the RST form, as shown in the block diagram of Fig. 3, where neglecting noise and disturbance, the following closed-loop transfer function in (15) can be derived in order to evaluate how the parameters of $Q(z^{-1})$ affect the system.

$$\frac{y(k)}{y_r(k)} = \frac{B(z^{-1})T(z^{-1})}{B(z^{-1})P(z^{-1}) + \Delta A(z^{-1})Q(z^{-1})}. \quad (15)$$

By keeping the cost polynomials $P(z^{-1}) = T(z^{-1}) = 1$, a constant $Q(z^{-1}) = \lambda$ modifies directly the characteristic polynomial of the closed-loop system, while the one shown in (14)

also allows modifications to the zeros, leading to the following closed-loop transfer function:

$$\frac{y(k)}{y_r(k)} = \frac{B(z^{-1})(1 - \gamma z^{-1})}{B(z^{-1})(1 - \gamma z^{-1}) + \lambda \Delta A(z^{-1})}. \quad (16)$$

Despite the coupling between zeros and poles related to the parameter γ in (16), using λ , the characteristic polynomial still could be manipulated to compensate undesired modifications to the poles, allowing a better bandwidth control of the closed-loop system [22,23]. To illustrate how the modified $Q(z^{-1})$ in (14) affects performance and robustness, in the following section, some robustness analysis results in terms of gain and phase margins, and also by a time-based robustness index found in [21,26,27], concerning PMM analysis for adaptive systems, are presented.

2.3. Robustness analysis

Instead of simply choose appropriate parameters for (14) based on trial and error through simulated output response alone, this paper also covers robustness analysis using sensitivity functions magnitude plots in the frequency domain. From those functions, important indices can be obtained to quantify the trade-off between robustness and performance to guarantee a suitable well-tuned controller. Such indices are: gain margin (GM), phase margin (PM), and maximum amplitude ratios [19].

The sensitivity function, $S_{\text{sen}}(z^{-1})$, characterizes the effect of a disturbance, $\xi(k)$, acting on the output of the control loop, whereas the complementary sensitivity, $T_{\text{com}}(z^{-1})$, is equivalent to the closed-loop transfer function of the control loop, i.e., from $y_r(k)$ to $y(k)$. These functions can be described by

$$S_{\text{sen}}(e^{j\omega T_s}) = \frac{1}{1 + G_c(z^{-1})G_p(z^{-1})}, \quad (17)$$

$$T_{\text{com}}(e^{j\omega T_s}) = \frac{G_c(z^{-1})G_p(z^{-1})}{1 + G_c(z^{-1})G_p(z^{-1})}, \quad (18)$$

where $G_p(z^{-1})$ is the process nominal model,

$$G_p(z^{-1}) = \frac{y(k)}{u(k)} = \frac{z^{-d}B(z^{-1})}{A(z^{-1})}, \quad (19)$$

and $G_c(z^{-1})$ is the incremental GMVC,

$$G_c(z^{-1}) = \frac{u(k)}{e(k)} = \frac{F(z^{-1})}{\Delta[B(z^{-1})E(z^{-1}) + C(z^{-1})Q(z^{-1})]}. \quad (20)$$

The maximum amplitude ratios of (17) and (18), respectively M_S and M_T , quantify the sensibility of the control loop to the excitation signal under consideration, which means that small M_S values make the system less sensible to $\xi(k)$, whereas M_T considers the influence of the reference signal, $y_r(k)$, and is equivalent to the amplitude of the resonant peak as well, that in general, is desirable to be kept small. In [19], the authors suggested that M_S and M_T values, respectively in the range of 1.2–2 and 1–1.5, give a suitable trade-off between robustness and performance. These two indices can be obtained from

$$M_S = \max_{\omega} |S(e^{j\omega T_s})|, \quad (21)$$

$$M_T = \max_{\omega} |T(e^{j\omega T_s})|. \quad (22)$$

As stated in the introductory part of this paper, a common practice of any robust unconstrained optimal design technique, is the usage of consolidated design criteria. From the results of Robust Control Theory, gain and phase margins have been vastly used to characterize robustness of control loops, and it is common sense, that $\text{GM} \rightarrow \infty$ and $\text{PM} = 60^\circ$, characterize

robust stability [28,29]. To obtain a softer compromise between robustness and performance, [19] suggests that GM and PM should be kept in the range of 1.7 dB–4 dB and 30° to 45° , respectively. Both indices can be obtained using the following equations from [19]:

$$\text{GM} \geq \frac{M_S}{M_S - 1} \quad (23)$$

$$\text{PM} \geq 2 \sin^{-1} \left(\frac{1}{2M_S} \right) \left(\frac{180^\circ}{\pi} \right) \quad (24)$$

$$\text{GM} \geq 1 + \frac{1}{M_T} \quad (25)$$

$$\text{PM} \geq 2 \sin^{-1} \left(\frac{1}{2M_T} \right) \left(\frac{180^\circ}{\pi} \right). \quad (26)$$

From Eqs. (23)–(26), two methods to obtain the robustness indices are given, each one based on $S_{\text{sen}}(z^{-1})$ and $T_{\text{com}}(z^{-1})$. These two methods are very effective in the design procedure, but not well suited for online robustness evaluation, even though any desktop computer can handle this task nowadays, as it is going to be shown on Section 5. However, a more suitable technique to evaluate robustness of self-tuning controllers online, is a time-based robustness index [21,26,27], given by

$$\text{RI}(k) = \frac{\left| \sum_{i=0}^k \hat{y}(i) - \sum_{i=0}^k y(i) \right|}{\left| \frac{\sum_{i=0}^k \hat{y}(i) \sum_{i=0}^k v(i)}{\sum_{i=0}^k w(i)} \right|}, \quad (27)$$

where $\hat{y}(k)$ is the estimated/simulated model output, $v(k) = T(z^{-1})y_r(k)$ and $w(k) = F(z^{-1})y(k)$. If $\text{RI}(k) < 1$, then the control loop is stable [26].

An example is shown next, to evaluate the robustness of GMV2DOF in comparison to the minimal order GMVC using the indices presented.

2.3.1. GMV2DOF robustness analysis example

Consider the reduced order damped pendulum model with polynomials

$$\begin{aligned} A(z^{-1}) &= 1 - 1.868z^{-1} + 0.9167z^{-2}, \\ B(z^{-1}) &= -0.0942 + 0.1368z^{-1}, \\ C(z^{-1}) &= 1, \quad d = 1, \end{aligned} \quad (28)$$

and sampling time (T_s) of 0.1 s. Using the model in (28), values for the parameters of GMV2DOF are obtained while focusing on GM between 1.7 and 4 dB and PM between 30° and 45° [19].

In face of what was seen in (16), the way how λ works for the modified $Q(z^{-1})$ in (14) is different from the common case and comparison between them should be done by means of best tuning regarding robustness and performance instead of using equally specified λ .

Designing the GMVC with common $Q(z^{-1})$ and with the modified $Q(z^{-1})$ to have almost the same robustness margins and performance, the results shown in Figs. 4 and 5 were obtained for the following parameters: $Q\{\lambda = 35\}$; $\text{mod}Q\{\lambda = 11, \gamma = 0.7\}$.

From Figs. 4 and 5 it is clear how both tunings look alike graphically and quantitatively, and viewing from this perspective, what is the big deal of using the modified structure? Suppose some non-minimum phase behavior have been forgotten when modeling the system and the B polynomial was in fact $B(z^{-1}) = -0.0942 + 0.1368z^{-1} + 0.11z^{-2}$. Using the same tuning simulated before, we see in Fig. 6 that despite the similarities found before, the modified structure still guarantees stability and servo tracking

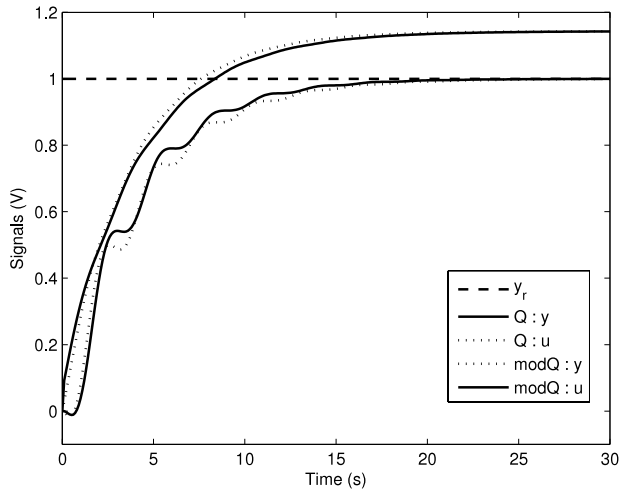


Fig. 4. Input and output signals for both cases of $Q(z^{-1})$.

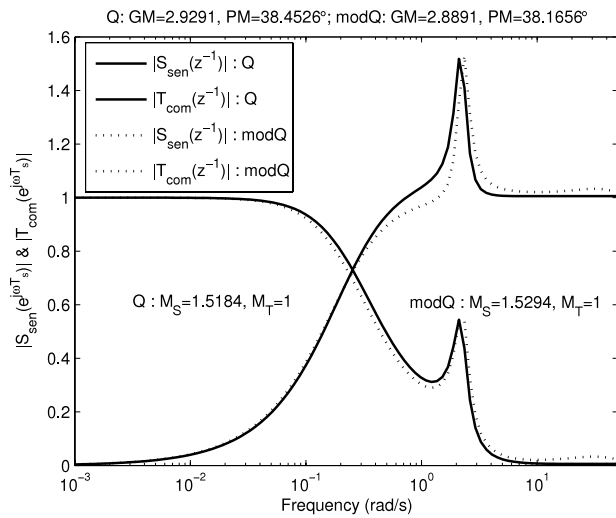


Fig. 5. Sensitivity and co-sensitivity functions for both cases of $Q(z^{-1})$.

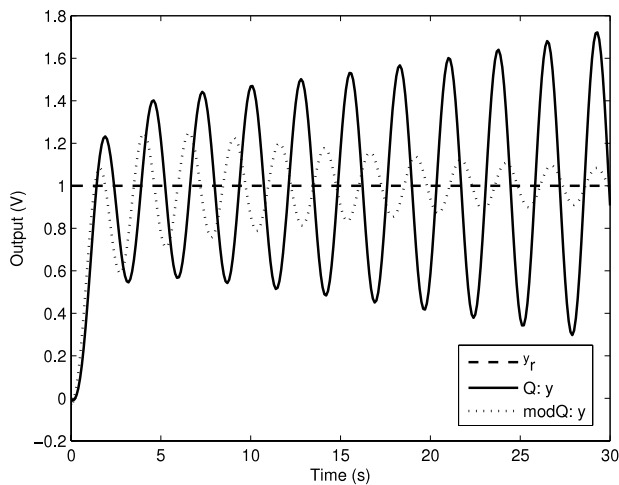


Fig. 6. Considering unmodeled non-minimum phase.

while the common structure fails, pointing to a better result of robustness to non-minimum phase systems for the GMV2DOF case in face of PMM.

The time-based robustness index of (27) was used to quantify how the GMV2DOF performed in the presence of PMM, with its

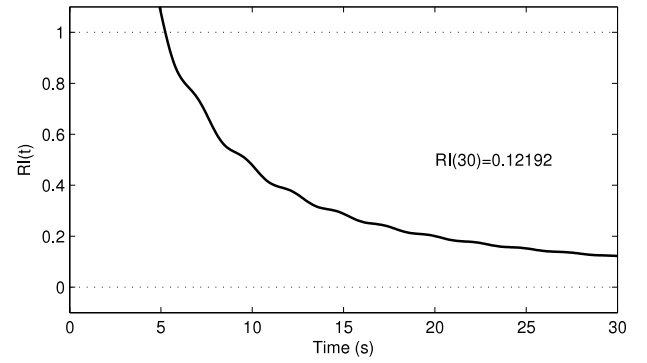


Fig. 7. Time-based robustness index results for the GMV2DOF facing PMM issues.

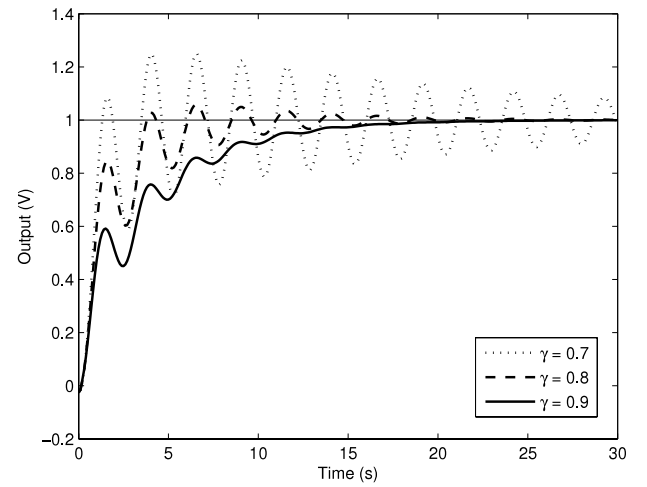


Fig. 8. Output responses for every γ tested in the GMV2DOF robustness analysis example.

Table 1
Robustness indices for different values of γ .

$\gamma = 0.7$	RI = 0.12192	GM = 2.8891	PM = 38.1656°
$\gamma = 0.8$	RI = 0.19831	GM = 3.9129	PM = 43.7048°
$\gamma = 0.9$	RI = 0.46572	GM = 5.9521	PM = 49.1642°

results being shown in Fig. 7, where the index $RI < 1$ shows that the control system is stable and its value at the end of the simulation [$RI(30) = 0.12192$] can be used to quantify robustness and performance. To emphasize this, Table 1 presents RI values, taken at $kT_s = 30$ s, with correspondent GM and PM for different values of γ , showing that small RI tends toward performance and, otherwise, toward robustness. The other parameter was kept at $\lambda = 11$ and the output responses for every case tested are shown on Fig. 8.

Tuning parameters used with the simulations in this section were obtained by trial and error while observing control and output signals along with the robust design criteria of [19]. From this experience, λ was found to be responsible for constrained control action and when increased, a more conservative control was observed and also an increase in gain and phase margins. Similar results were found when increasing γ , but its role is more directly related to deal with oscillatory modes, leading to more freedom when trying to establish a good trade-off between performance and robustness.

3. GMV2DOF with direct STC

All effort till this section was headed toward guaranteed robustness when designing based on linear reduced-order models. Now,

the main goal is to embed that previously acquired knowledge inside an adaptive structure that provides an updated control law and consequently higher fidelity to manage the actual state of the process by adapting directly the controller in order to avoid process identification errors when dealing, for example, with complex and noisy industrial environments. Since GMVC is a model based control structure, an adaptive approach guarantees redundancy to robustness.

As stated before, the STC in this work is used in the direct form, that is, estimates directly the controller parameters instead of identifying the process model and then compute the control law [1]. So, by taking the generalized output from Eq. (2) and shifting it back d -steps behind, keeping $P(1) = T(1) = 1$ and rearranging the modified $Q(z^{-1})$ in the equation, gives:

$$\bar{\phi}(k) = [1 - \gamma z^{-1}][y(k) - y_r(k)] + \lambda \Delta u(k - d) \quad (29)$$

where $\bar{\phi}(k) = (1 - \gamma z^{-1})\phi(k)$. By comparison of (29) with the RST mapping seen on Fig. 3, gives:

$$\bar{\phi}(k) = [1 - \gamma z^{-1}][S(z^{-1})y(k) - T_{\text{RST}}(z^{-1})y_r(k)] + \bar{R}(z^{-1})\Delta u(k - d) \quad (30)$$

where $\bar{R}(z^{-1})$, $S(z^{-1})$, $T_{\text{RST}}(z^{-1})$, respectively have the following order dependence from $B(z^{-1})E(z^{-1}) + C(z^{-1})Q(z^{-1})$, $F(z^{-1})$ and $z^d C(z^{-1})T(z^{-1})$, leaving the controller parameters in $\hat{\theta}$ (e.g., based on a second-order CARIMA design model with $d = 1$, $C(z^{-1}) = 1$) to be estimated by the RLS algorithm using the measurements vector ϕ_{LS} as in

$$\hat{\theta} = \begin{bmatrix} s_0 \\ s_1 \\ s_2 \\ t_0 \\ \bar{r}_0 \\ \bar{r}_1 \end{bmatrix}, \quad \phi_{LS} = \begin{bmatrix} (1 - \gamma z^{-1})y(k) \\ (1 - \gamma z^{-1})y(k - 1) \\ (1 - \gamma z^{-1})y(k - 2) \\ -(1 - \gamma z^{-1})y_r(k) \\ \Delta u(k - d) \\ \Delta u(k - 1 - d) \end{bmatrix}. \quad (31)$$

The STC RLS algorithm workflow is:

1. Give initial values for $\hat{\theta}$ and for the covariance matrix $P_{\text{cov}}(k) = xI_{6 \times 6}$, where x is some appropriate startup value;
2. Acquire ϕ_{LS} and compute Eq. (29);
3. Compute the estimator error (e_{est}) and gain (K_{est}):

$$e_{\text{est}}(k) = \bar{\phi}(k) - \phi_{LS}^T(k)\hat{\theta}(k),$$

$$K_{\text{est}}(k) = \frac{P_{\text{cov}}(k)\phi_{LS}(k)}{[1 + \phi_{LS}^T(k)P_{\text{cov}}(k)\phi_{LS}(k)]};$$

4. Update the parameters vector and the covariance matrix using

$$\hat{\theta}(k) = \hat{\theta}(k - 1) + K_{\text{est}}(k)e_{\text{est}}(k),$$

$$P_{\text{cov}}(k) = P_{\text{cov}}(k - 1) - K_{\text{est}}(k) \times [1 + \phi_{LS}^T(k)P_{\text{cov}}(k - 1)\phi_{LS}(k)]K_{\text{est}}^T(k);$$

5. Calculate $\Delta u(k)$ in (30) for $\bar{\phi} \rightarrow 0$ and apply the updated control signal to the plant: $u(k) = u(k - 1) + \Delta u(k)$;
6. Return to step 2.

4. GMV2DOF with Fuzzy Scheduling

In this section, the self-tuning derived is aimed at adapting the parameters of the modified $Q(z^{-1})$ with a Mamdani Fuzzy Scheduler [6,30], so that robustness design criteria hold between some desired region [23]. The criteria evaluates together both gain and phase margins, studied earlier in Section 2.3, and the behavior of the process output variation in order to decide how to tune the parameters of the controller.

Table 2
Base of rules for Fuzzy Scheduling λ .

	LP	IP	HP
LG	SD	D	I
IG	D	I	UD
HG	I	UD	PD

The expertise applied into the Fuzzy Scheduling was granted by many different analysis made in this paper, from analytical to physical behavior of the processes involved, and also, trying to generalize its usage to other classes of plants. The main idea is linking λ to handle tracing ideal gain and phase margins while γ handles transitory behavior in order to increase performance. In this sense, the problem was divided into two fuzzy systems for each parameter of GMV2DOF, as described next.

4.1. Fuzzy Scheduling λ

Fuzzy Scheduling λ is done based on gain and phase margins. It is necessary to use online identification (by RLS algorithm) of the process, to have in every sample time, an updated second order model to evaluate robustness using the sensitivity functions shown in (17) and (18). After that, gain and phase margins are obtained using (23) and (24), and are sent as input variables to the Fuzzy Scheduling system.

Gain margin universe of discourse is evaluated by three membership functions (MF), named as: Low Gain (LG), Ideal Gain (IG) and High Gain (HG). They are respectively represented by trapezoidal-triangular-trapezoidal functions. The same is done for phase margin, naming the MFs as: Low Phase (LP), Ideal Phase (IP) and High Phase (HP). Those MFs are placed in the following manner:

- LG: $\{-\infty, 2 \text{ dB}, 3 \text{ dB}\}$, IG: $\{2 \text{ dB}, 3 \text{ dB}, 4 \text{ dB}\}$, HG: $\{3 \text{ dB}, 4 \text{ dB}, +\infty\}$;
- LP: $\{-\infty, 30^\circ, 37.5^\circ\}$, IP: $\{30^\circ, 37.5^\circ, 45^\circ\}$, HP: $\{37.5^\circ, 45^\circ, +\infty\}$.

For the output, λ , five triangular MFs are used and are equally spaced into λ universe of discourse. Chosen MF centers are $\{7, 9, 11, 13, 15\}$, respective to Poorly Damped (PD), Un-Damped (UD), Ideal (I), Damped (D) and Strongly Damped (SD).

The base of rules designed is shown in Table 2, where outer cells represent inputs MFs and inner cells the corresponding activated output MF.

4.2. Fuzzy scheduling γ

The scheduling of γ is mainly related on constraining absolute values of output variations (OV) during transitory ending stages and letting it go more freely during steady-state stages. For this purpose, $\text{abs}(y(k) - y(k - 1))$ is sent to the fuzzy scheduler as its input variable, where three MFs are evaluated: Low OV, Ideal OV and High OV, in trapezoidal-triangular-trapezoidal form, respectively spaced as: $\{-\infty, 0.01, 0.02\}$, $\{0.01, 0.02, 0.03\}$, $\{0.02, 0.03, +\infty\}$.

This fuzzy system has a direct mapping to three triangular MFs regarding γ as output: un-damped, ideal and damped, respectively centered at $\{0.6\}$, $\{0.7\}$ and $\{0.8\}$. So, if OVs become small, this fuzzy scheduler decreases the value of γ , tuning the GMV2DOF in an alert condition to set-point changes, letting the change occurs more freely. Otherwise, the scheduler increases γ in order to reduce oscillations when the output gets closer to the desired reference.

The defuzzification method used in both Fuzzy Scheduling schemes is center-average [31].

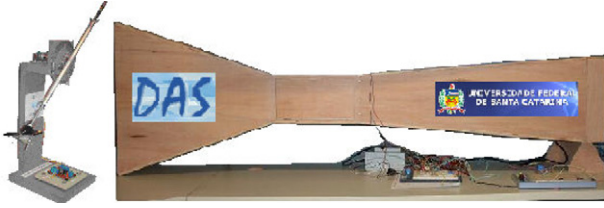


Fig. 9. Damped pendulum (left) and wind tunnel (right).

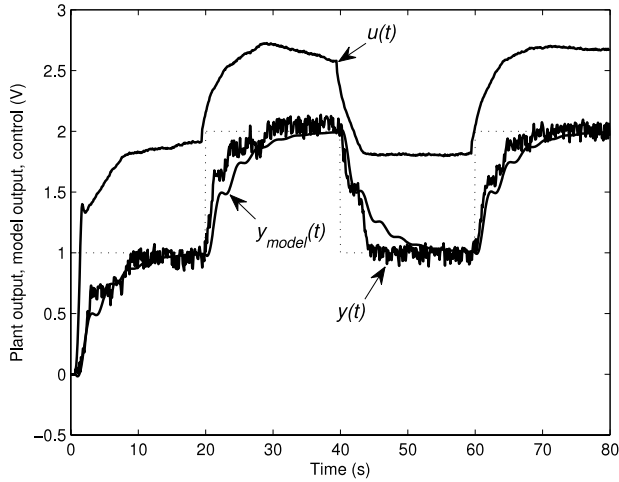


Fig. 10. Control and output response of the STC GMV2DOF with the damped pendulum.

5. Experimental setup and results

The experimental setup is comprised of two different processes, a damped pendulum and a wind tunnel, as shown in Fig. 9. The damped pendulum takes position measurements from a potentiometer attached to its axis and is controlled by a propulsion system that includes a DC motor and propeller. The wind tunnel measures airstream velocity by means of a differential pressure sensor, the MPX5010DP, connected to a pitot-static (Prandtl) tube and is controlled by a RC aircraft propulsion system also composed of a DC motor and propeller. For both processes, input and output signals are measured in Volts (V) in the range of 0–5 V.

Fig. 10 presents the results of the STC GMV2DOF applied to the real damped pendulum process. In the same figure, the nominal model output response is also presented for comparison and evaluation of the RI index, where the nominal model response was simulated using GMV2DOF alone, i.e., without STC. Controller parameters used were based on the robust design from Section 2.3.1, where $\lambda = 11$, $\gamma = 0.7$, and even being the same in the real process experiment and in the nominal model simulation, the STC performed slightly better with respect to the speed of response, probably due to performance favorable mismatches of plant and model, well compensated by the STC algorithm that kept the output as close as possible to design specifications ($GM = 1.7\text{--}4\text{ dB}$, $PM = 30^\circ\text{--}45^\circ$). The RI index of the experiment is presented in Fig. 11, ensuring guaranteed stability of the control loop and, STC parameters adaptation in RST form are shown in Fig. 12.

Still with the damped pendulum process, Fig. 13 illustrates the results of the Fuzzy Scheduled GMV2DOF adaptive strategy. Similar to what was done in the STC experiment, in this new case, the experimental results are compared to the nominal model results in order to verify if the plant performs as expected by design and also to generate the RI index shown in Fig. 14. Since the Fuzzy Scheduling strategy adopted was headed toward increased

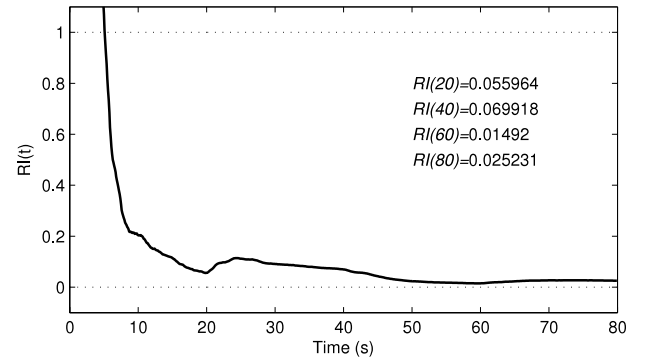


Fig. 11. Time-based robustness index RI of the STC GMV2DOF with the damped pendulum.

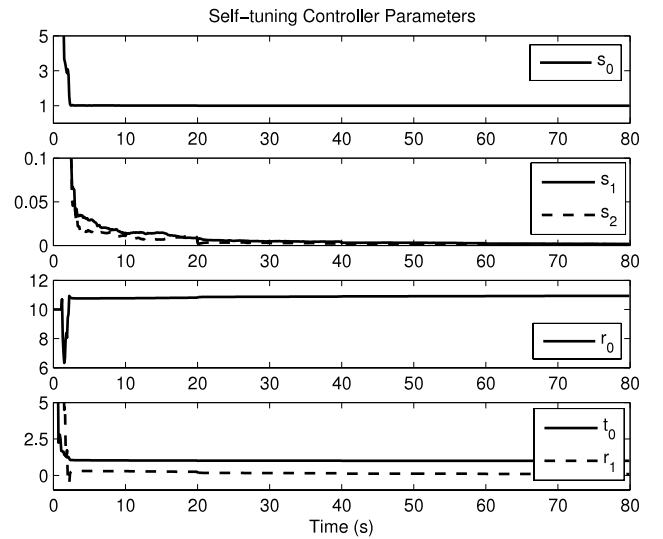


Fig. 12. STC GMV2DOF, in RST form, parameters adaptation over time.

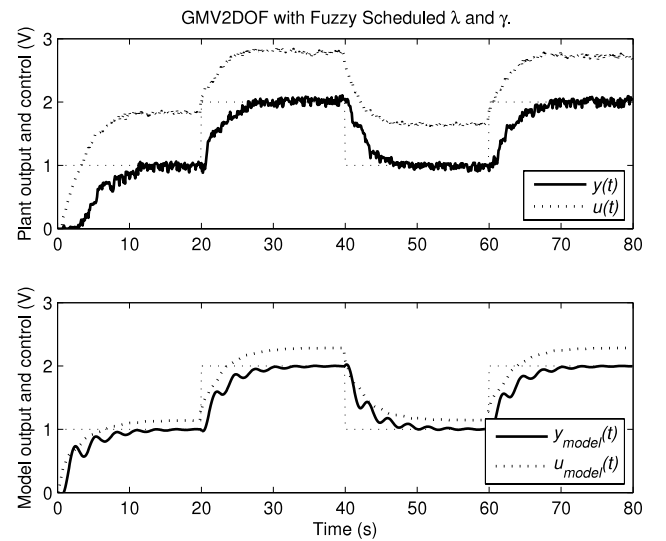


Fig. 13. Damped pendulum process and model responses with the Fuzzy Scheduled GMV2DOF.

performance during setpoint changes and increased robustness in steady-state, the speed of response was expected to outperform the STC case, but it also represented well the simulations with the nominal model and gave a good RI index in comparison to the STC case of Fig. 11. The flexibility granted to λ and γ , also the more

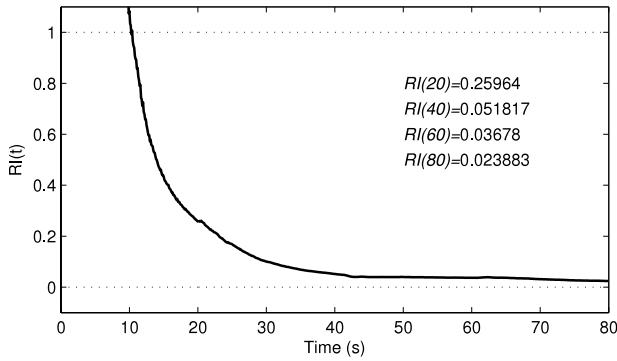


Fig. 14. Time-based robustness index RI of the Fuzzy Scheduled GMV2DOF with the damped pendulum experiment.

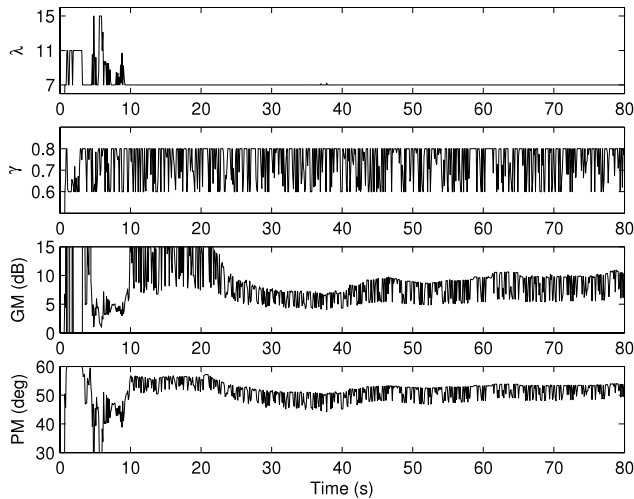


Fig. 15. Parameters adaptation of the Fuzzy Scheduled GMV2DOF based on GM and PM observations over time.

accurate – to the time of the experiment – online identified model, allowed increased GM and PM as depicted by Fig. 15.

As self-tuning strategies, both STC GMV2DOF and GMV2DOF with Fuzzy Scheduling, should manage to self-tune even when facing a new *unknown* process. In that case, the new *unknown* process used as example in this research, was the wind tunnel, which did not happen to have any pre-design stage. All the settings of the experiments with the damped pendulum process were used to control the wind tunnel. Results are shown in Figs. 16 and 17, respectively for the STC and Fuzzy Scheduled cases. Despite the extremely noisy signals verified in the figures, due to the fact that the airstream inside the tunnel is turbulent and the pressure sensor signal used had to be highly amplified, both adaptive techniques handled well the task at hand with very similar results. The more significant difference is related to the noisy control signal of the Fuzzy Scheduling case, due to its constant and abrupt fast switching of GMV2DOF parameters under extremely noisy conditions as presented here. However, it does not mean that it is unfeasible under noisy conditions, but an output filter design for GMV $P(z^{-1})$ polynomial is recommended. Here, it was set to $P(z^{-1}) = 1$.

Although the adaptive techniques presented handled well an *unknown* process, it was mere coincidence due to similarities between the damped pendulum and wind tunnel dynamics, input/output operational range, and also the intrinsic robustness of the GMVC structure. Special care should be taken to the process DC gain, because suitable selection of the control weighting factor λ is directly related to it. For the STC case, this is an important remark, since the λ and γ values are fixed, and if wrongly selected, control

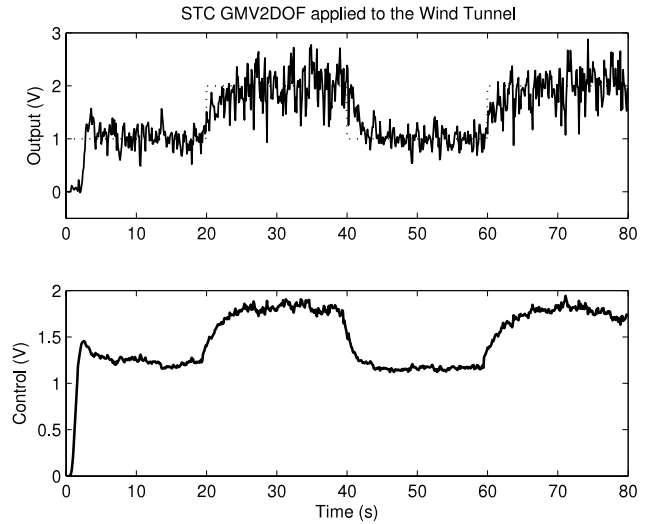


Fig. 16. STC GMV2DOF applied to the wind tunnel process.

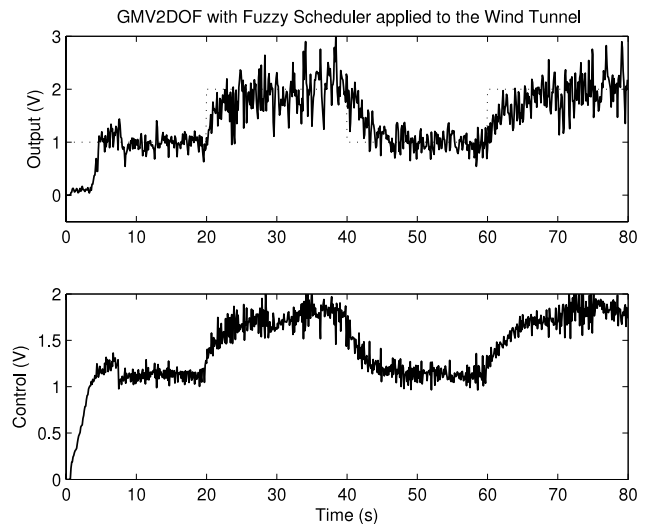


Fig. 17. GMV2DOF with Fuzzy Scheduled λ and γ applied to the wind tunnel process.

saturation and instability issues might happen. For the Fuzzy Scheduling case, variables universe of discourse range could be expanded and a conservative startup adopted in order to overcome DC gain difference issues, which makes this adaptive technique more flexible to processes variations and to its wider applicability.

6. Conclusions

This work was in no way aimed to do comparisons between both techniques studied, but it lay in a position of pre-studying them in order to merge knowledge acquired from both. Depicting some pros and cons, the STC studied is awarded for the fact that it estimates directly the controller, avoiding process identification problems that could mislead the controller design. In contrast, for the Fuzzy Scheduling developed, process online identification was used in order to do frequency response analysis in real-time to tune the controller parameters, but is computationally burdensome. However, nowadays any desktop computer can handle this task and, on the other hand, there are pros to the Fuzzy Scheduling when facing the need to adapt the control system to an unknown new process, where the STC GMV2DOF would need a pre-design stage in order to obtain suitable fixed parameters to guarantee robustness and desired performance.

Performance criteria, commonly expressed in terms of rise time, damping ratio, maximum overshoot and settling time, was adopted in a different way in this work, based on gain and phase margins specification that indirectly reflects those commonly used performance criteria. Beyond that, this form of analysis considers the trade-off with robustness and is a standard in the Robust Control Theory.

Both adaptive techniques presented in this work are partially functioning as supervisory and decision systems, and could be applied to other types of controllers with similar characteristics with respect to GMVC, such as PI (Proportional-Integral) and IMC (Internal Model Control) controllers.

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